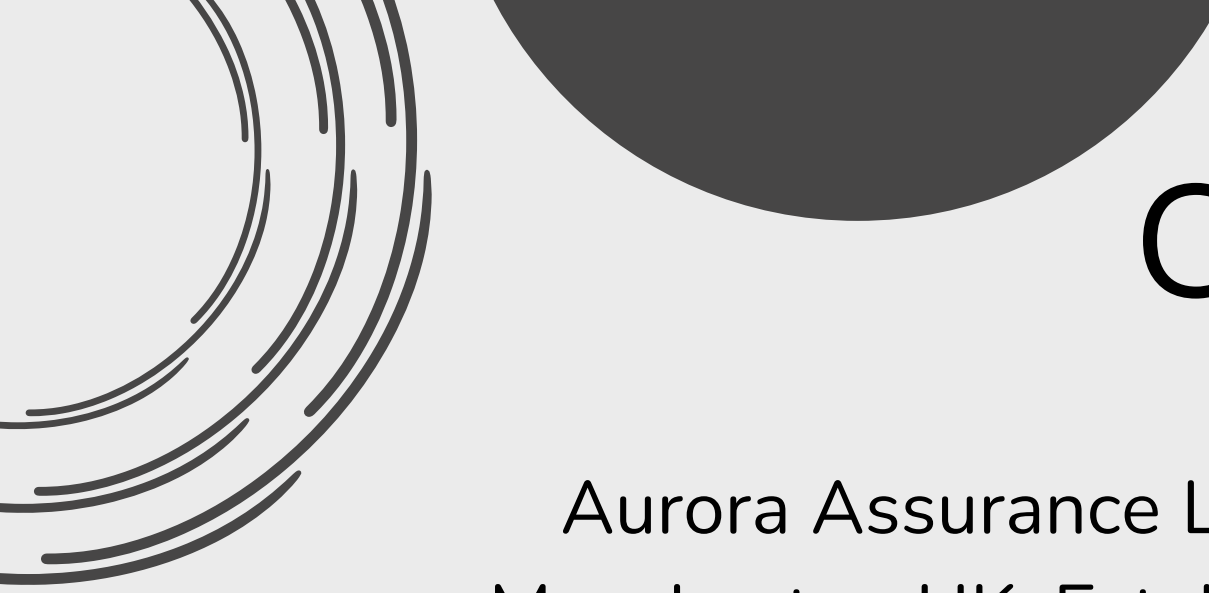




AURORA ASSURANCE LTD

Fraud Risk Intelligence Analytics

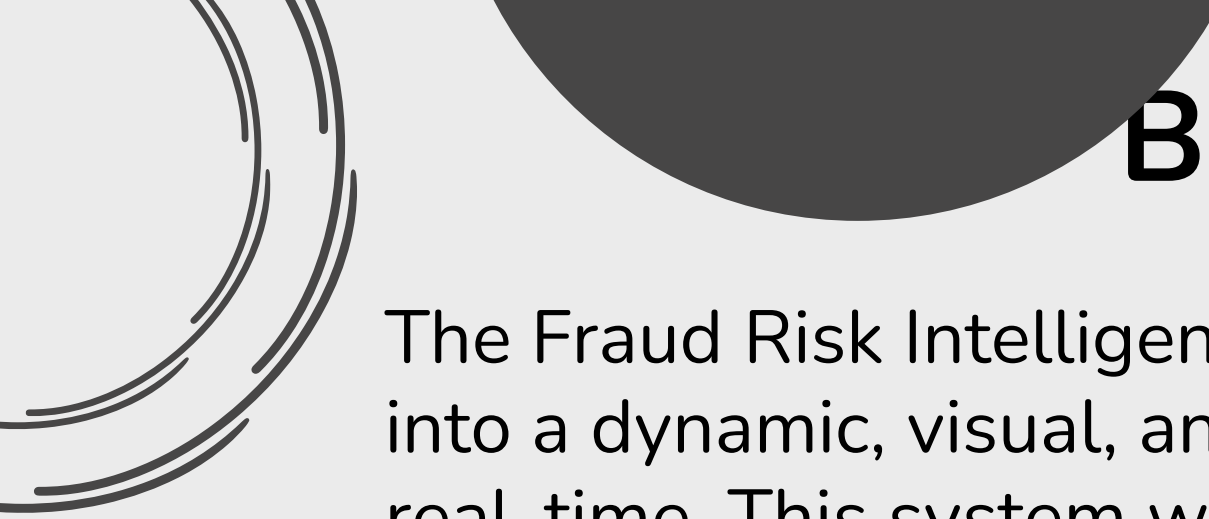




Company Overview

Aurora Assurance Ltd. is a mid-sized insurance provider based in Manchester, UK. Established in 2003, the company has grown from a small regional insurer into a recognized leader in the UK and emerging European markets. Aurora offers a comprehensive range of insurance products, including auto, home, travel, and business insurance, and currently serves over 500,000 active policyholders.





Business Challenge / Project Aim

The Fraud Risk Intelligence Analytics project aims to transform Aurora's claims data into a dynamic, visual, and interactive reporting system that detects fraud patterns in real-time. This system will directly address the challenges of fraud detection and operational inefficiencies.

Aurora Assurance Ltd. is facing a growing challenge in managing fraudulent claims, with a 14% increase in flagged fraudulent claims over the last two years. The key issues contributing to this challenge include:

1. No Real-Time Fraud Monitoring System: Fraudulent claims are often detected only during post-settlement audits, which causes significant delays in identifying suspicious activity.
 2. Fragmented Data Sources: Claims data is dispersed across multiple systems, making it difficult to conduct a holistic analysis and spot patterns of fraud effectively.
 3. Delayed Detection: Fraudulent claims slip through the initial review process, resulting in substantial revenue leakage due to overpayments, underpayments, and unnecessary payouts.
 4. Customer Trust Risk : Slow investigations frustrate genuine claimants and harm brand reputation, making it crucial for Aurora to act quickly to improve its fraud detection systems.
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Tailored Analysis

Fraud Detection & Risk Insights

1. Which claims are most frequently flagged as potential fraud?
2. Do regions differ in fraud incidence rates?
3. Are weekend submissions more likely to be fraudulent?
4. Are fraud flagged claims linked to short/ long term customers?
5. Which claim type has the highest fraud score?

Operational Efficiency & Claim Processing

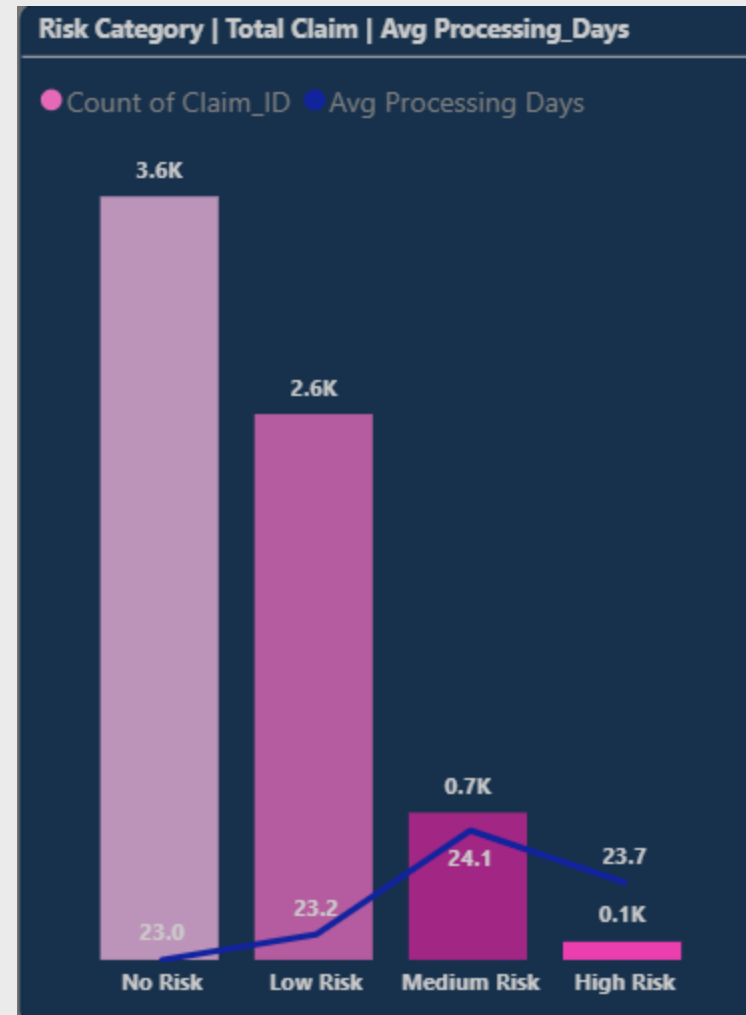
1. Do fraudulent claims usually take longer to resolve?
 2. What is the average claim processing time?
 3. Which claim type has the highest claim amount
 4. What is the claim trend over time?
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KPIs

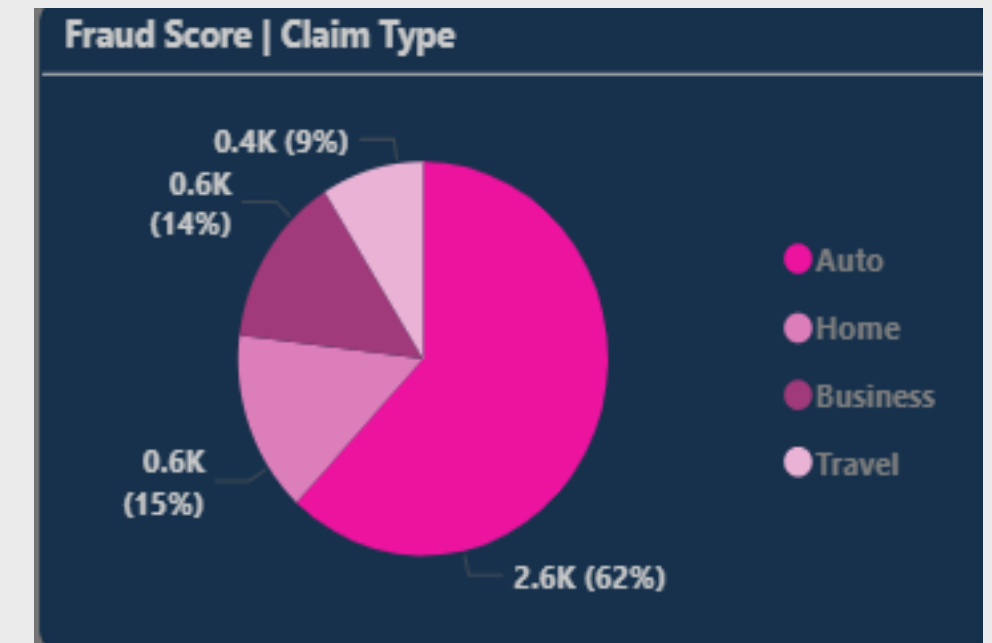


There was a total of 7,000 claims received from 3,000 customers within 6 regions. These claims were handled by 300 Adjusters over an average processing time of 23.2 days. Out of the 7,000 claims received, 2.34% were flagged as possible duplicate and 8.17% were flagged as suspicious claims. 36.65% of the total claim amount were flagged as high value claim.

Fraud Detection & Risk Insights

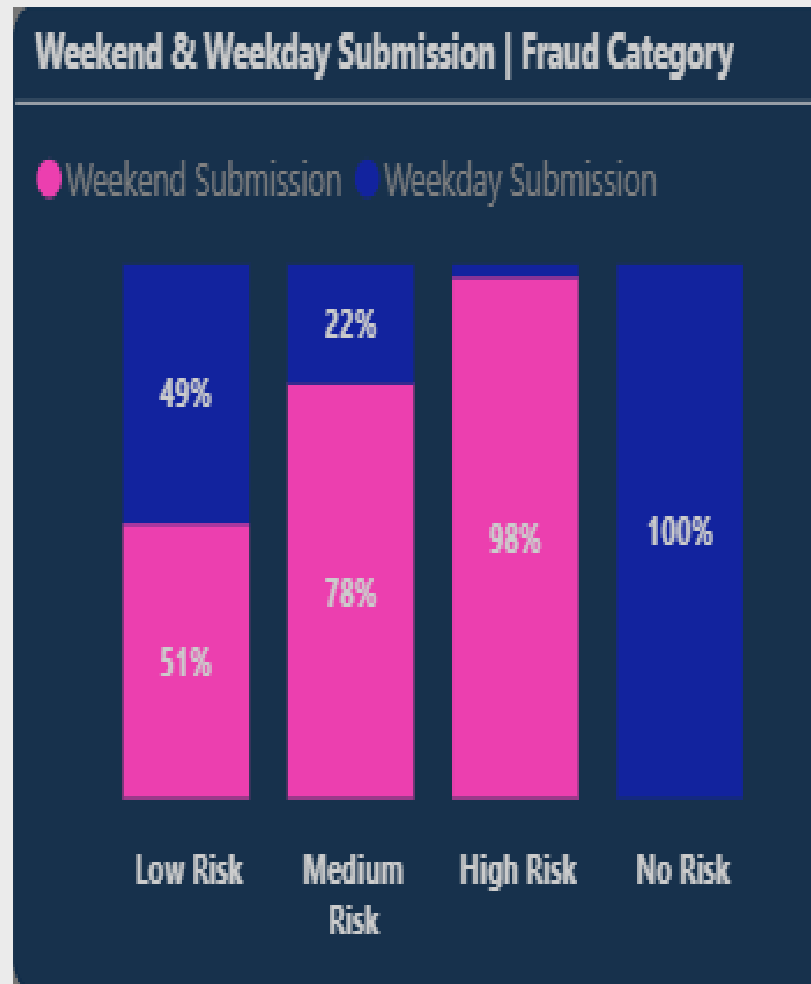


Out of the total claims of 7,000 about 3,600 claims had no risk (risk score of zero) which is more than half of the total claims and only about 1% of the total claims had high risk. However, the average processing time across the various categories remain in the same range indicating that adjusters may be spending more time on claims that has no risk as opposed to high-risk claims or fraudulent claims which frustrates genuine claimants and harm brand reputation.

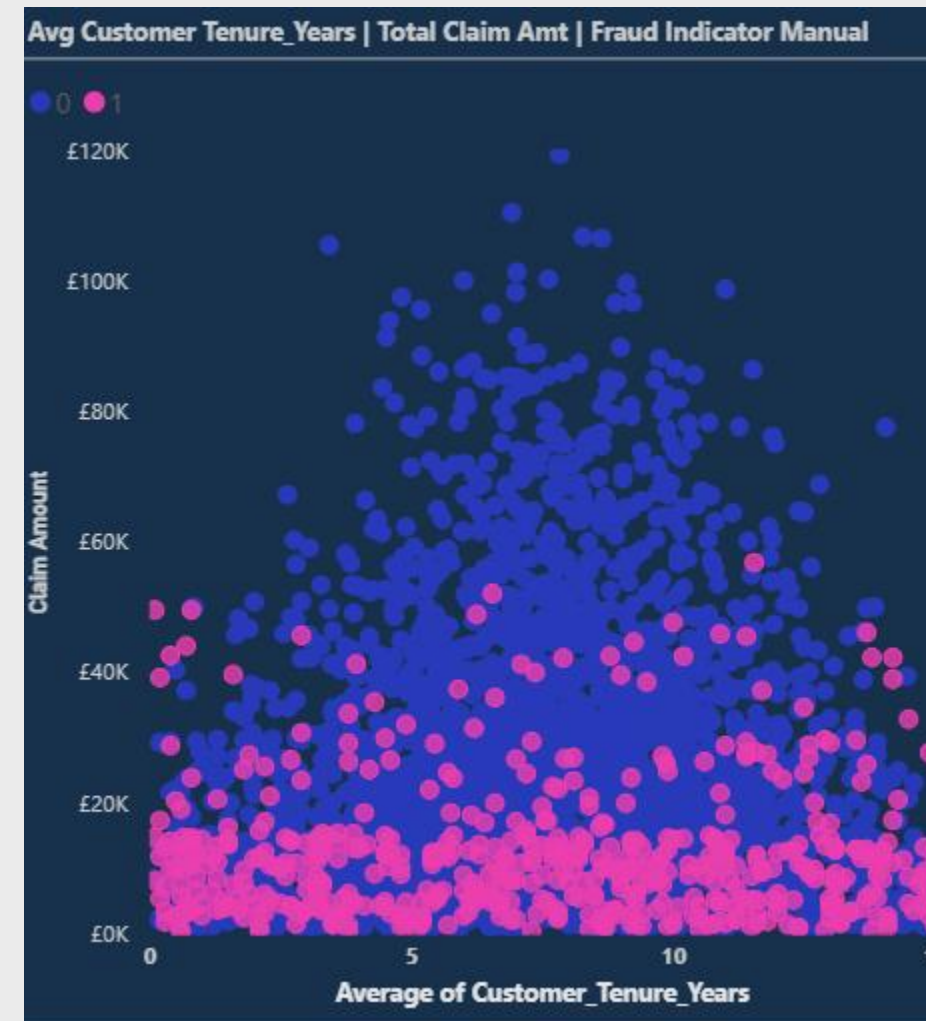


Out of the 4 claim types, Auto had the highest score for fraudulent claims (62%) indicating these claims are more likely to be fraudulent. Travel however, had the least score (9%) indicating these claims are less likely to be fraudulent.

Fraud Detection & Risk Insights Cont.



Comparing the Weekday/Weekend submissions with the risk categories, weekend submissions can be deemed to be highly fraudulent as 98% of the weekend submissions are marked as high risk. On the other hand, 100% of the claim submissions on weekdays had no risk (risk score of zero) which means all the weekend submissions had some level of risk. This clearly indicate that weekend submissions are generally fraudulent in nature compared to weekday submissions.



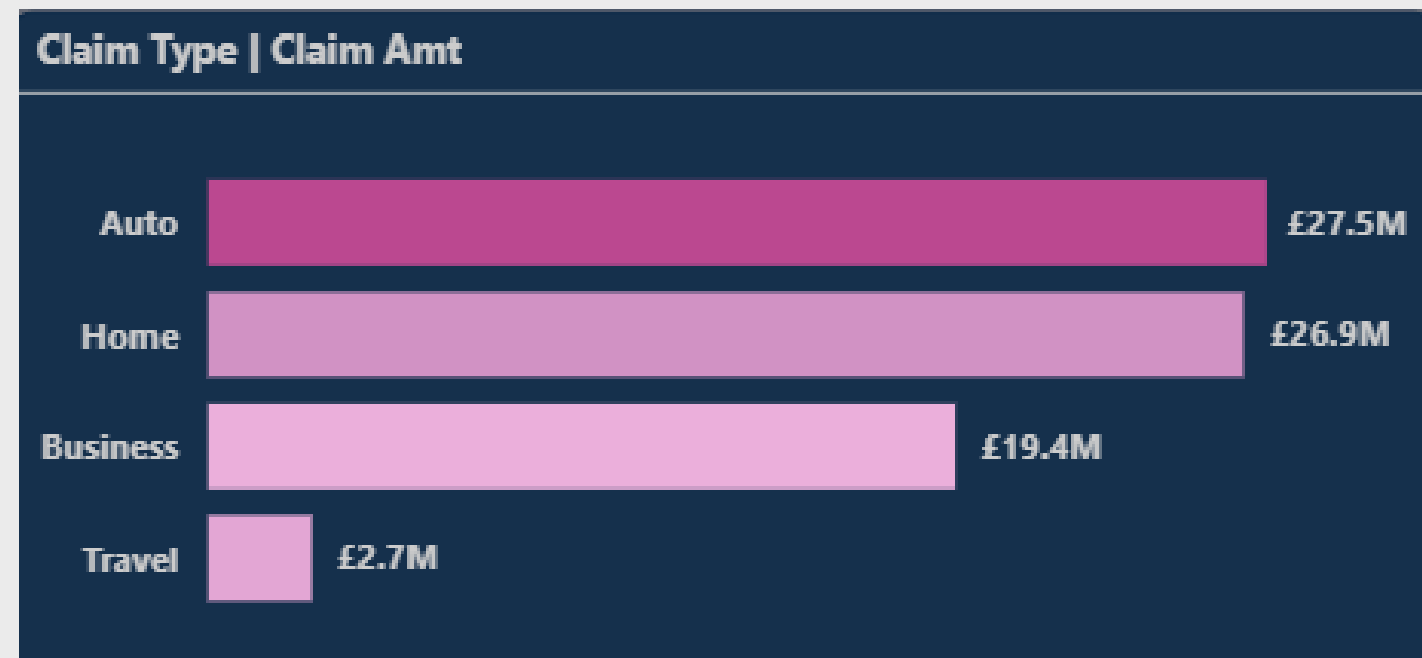
Claims with fraud indicator manual of 1 (suspicious claims) are fairly distributed across the average customer tenure years. This means fraudulent claims are not directly linked to short term customers but cuts across both short and long-term customers. The fraudulent claims are crowded at the bottom between £0 and £20K claim amount therefore claims ranging between these amounts need to be investigated critically.



Recommendation

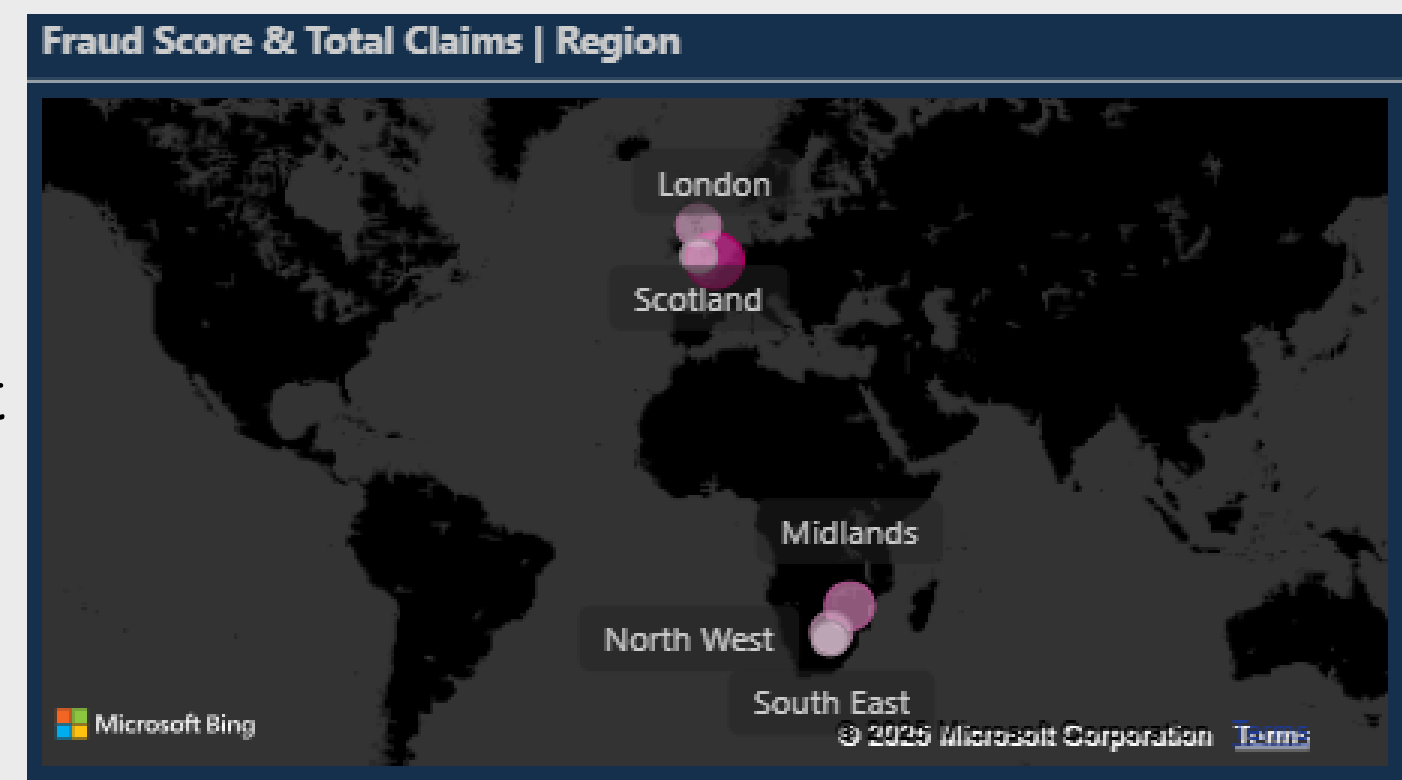
- ❖ Automate low-risk claims and redeploy adjuster time to medium/high risk.
- ❖ Strengthen fraud monitoring for Auto claims as they are the main fraud driver.
- ❖ Flag weekend submissions as higher risk; these require secondary review before payout.

Operational Efficiency & Claim Processing



Auto has the highest total claim amount over the period. Auto as well had the highest fraud score of 62% which may indicate that a high portion of the total claim amount may be fraudulent. Similarly, home claims had the second highest fraud score (15%), and it is the claim with the second highest total claim amount following closely from Auto claims.

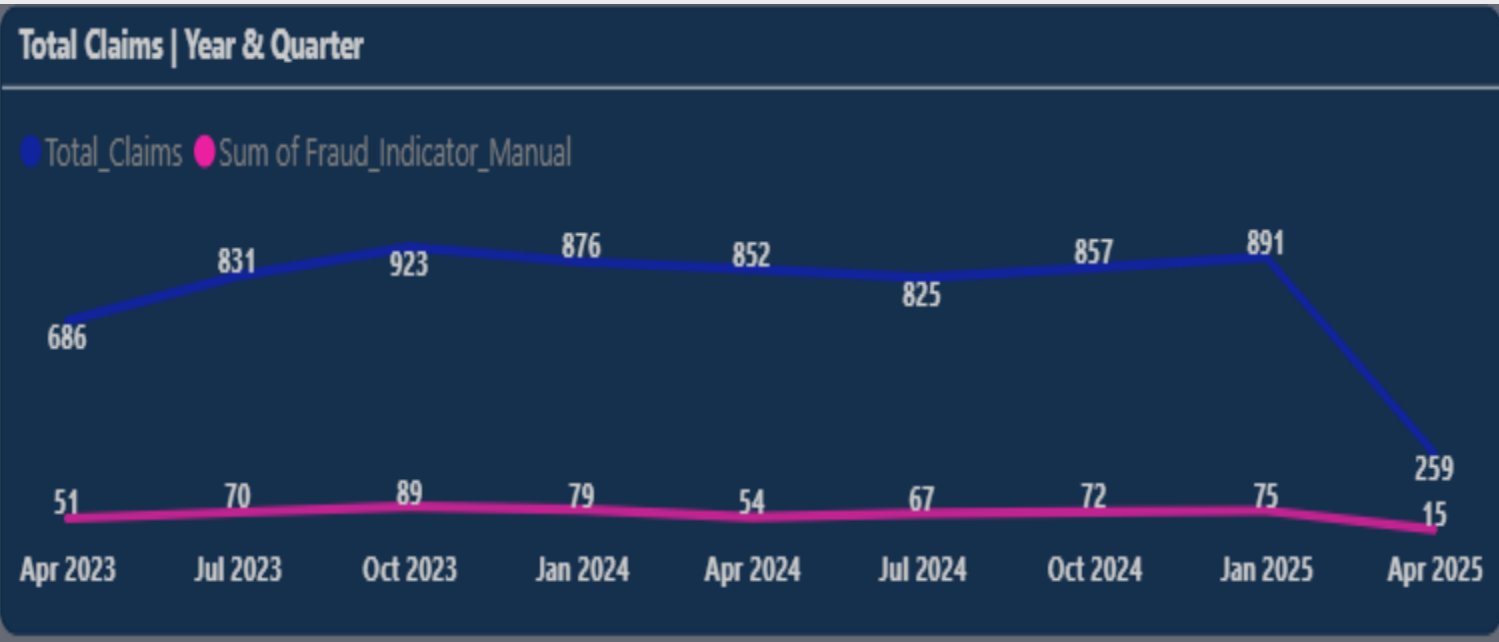
This shows the total claims and fraud risk score for each region. The deeper the colour, the higher the fraud risk score. London is the region with the highest number of claims and fraud score. Midlands had the second highest score.



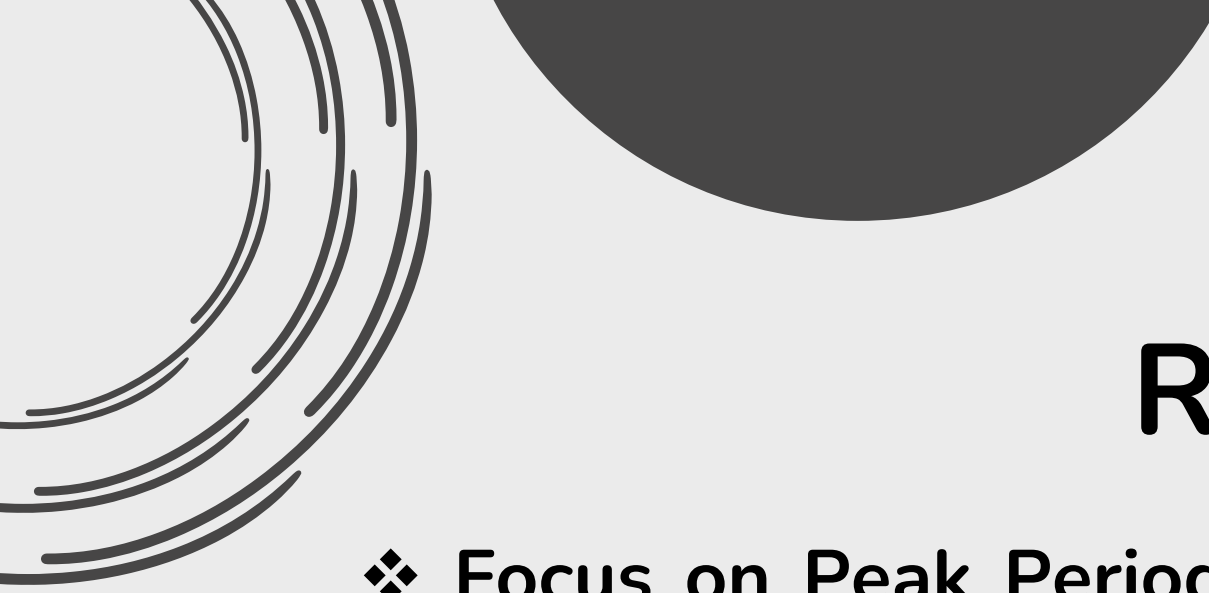
Operational Efficiency & Claim Processing Cont.

TOP 10 Adjusters				
AdjusterID	Total_Claim_Amt	Average of Processing_Time_Days	Total_Claims	Sum of Fraud Score
64	£518,534.53	23.07	29	24
176	£512,963.41	24.74	38	26
142	£495,810.70	20.72	39	26
233	£492,399.97	18.51	37	17
167	£475,069.09	25.44	32	23
162	£437,930.31	23.80	40	26
158	£424,928.05	22.10	31	21
272	£417,216.13	22.46	35	22
195	£416,578.71	21.92	24	16
67	£414,560.25	24.22	32	24

This shows the top 10 Adjusters based on the total claim amount. The adjusters with the highest fraud score of 26 handled the most claims. Adjusters who processed claims faster e.g., 233 do not necessarily have higher fraud score showing efficiency can coexist with accuracy.



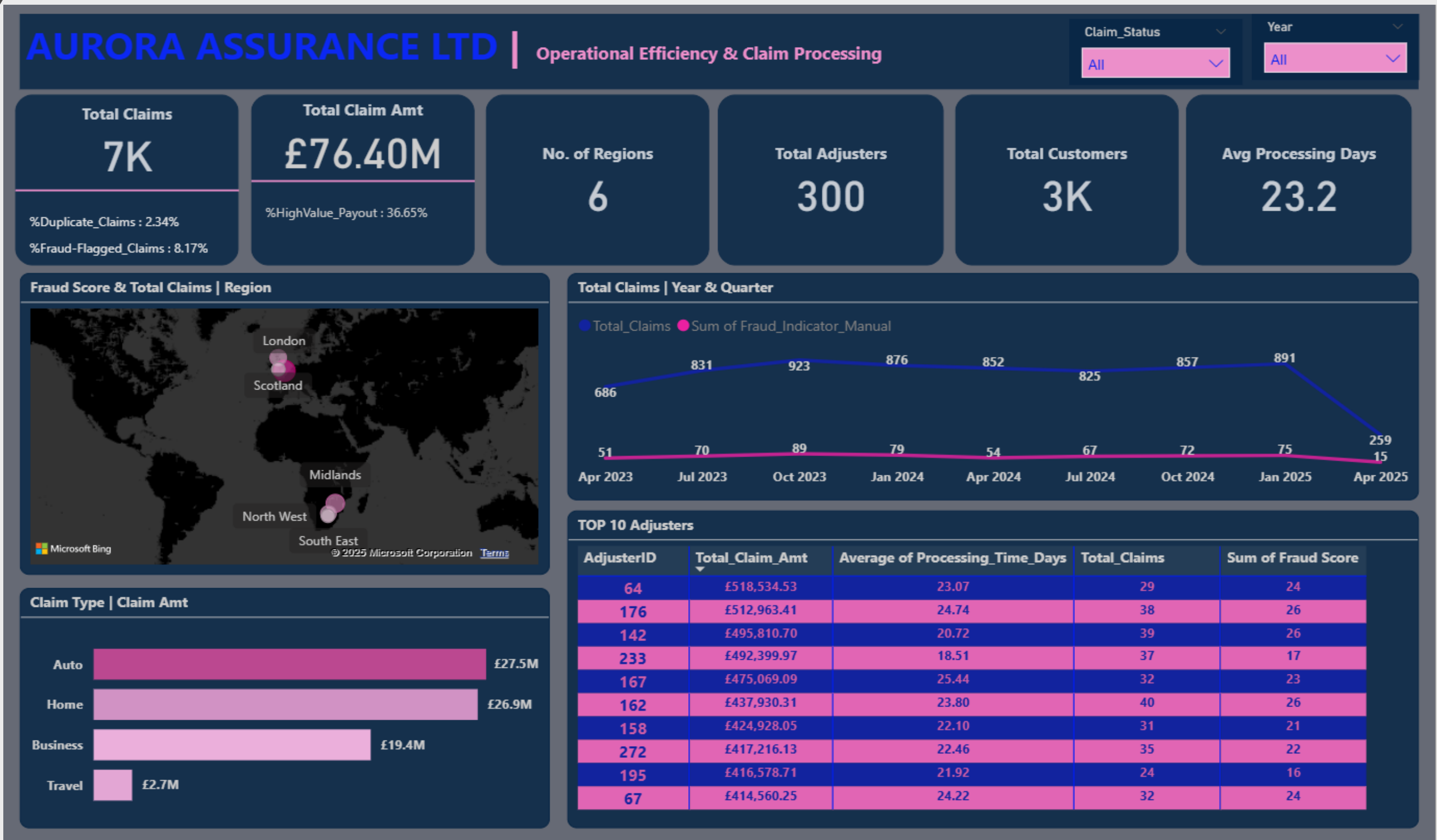
This shows the trend of claims over time as well as the fraud indicator score. The claims seem to be steady in the same range across the various quarters of the years. Total claims together with fraud risk peaked in October 2023.



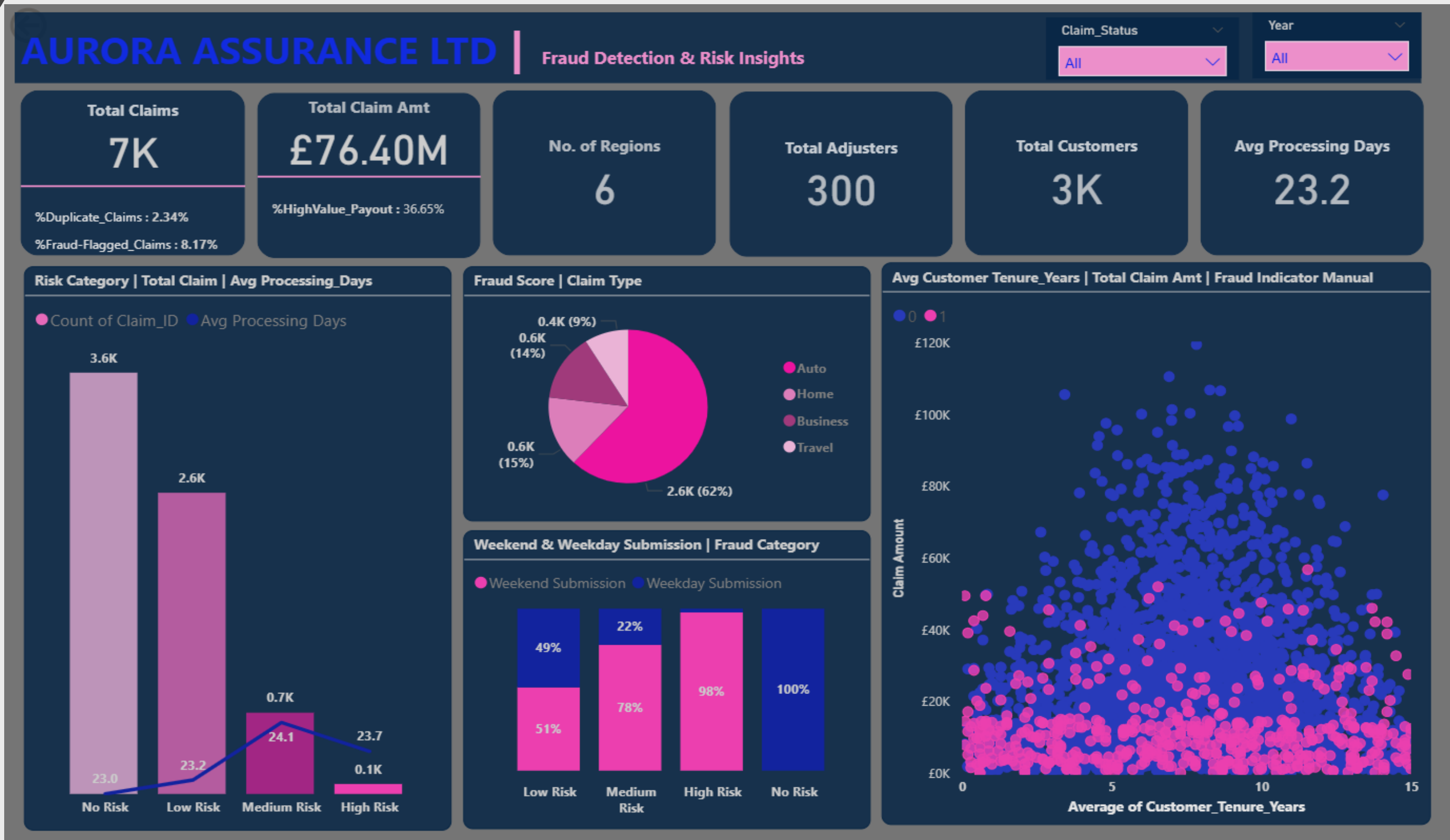
Recommendation

- ❖ **Focus on Peak Periods:** During claim surges (Oct 2023, Jan 2025), fraud monitoring should be tightened because fraudsters may exploit busy periods when oversight is weaker.
 - ❖ **Leverage Best Practices from Low Fraud Adjusters:** Adjuster 195 has a low fraud score (16) and reasonable claim amounts. Their approach could be studied and shared as a benchmark for fraud prevention.
 - ❖ **Fraud Prevention Training:** Provide targeted training for adjusters with higher fraud scores to improve red-flag detection.
 - ❖ **Investigate High Fraud Score Adjusters:** Their processes, claim documentation, and decision-making should be reviewed for risk of approving fraudulent claims.
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Dashboard 1



Dashboard 2



Fraud Score | Claim Type

Claim Type	Fraud Score	Percentage
Auto	2.6K	62%
Home	0.6K	15%
Business	0.6K	14%
Travel	0.4K	9%

Weekend & Weekday Submission | Fraud Category

Fraud Category	Weekend Submission	Weekday Submission
Low Risk	51%	49%
Medium Risk	78%	22%
High Risk	98%	2%
No Risk	100%	0%

Avg Customer Tenure_Years | Total Claim Amt | Fraud Indicator Manual

Avg Customer Tenure_Years	Total Claim Amt	Fraud Indicator Manual
0	£0K	0
5	£40K	1
10	£80K	1
15	£120K	1



THANK YOU

